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Subject: ECE3004

Microprocessors and Microcontrollers

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Course type: LTP

Credits: 4

Max 5 sessions

COMPUTER SYSTEM COMPONENTS

- **Memory**

  Stores instructions and data

- **Input/Output**
  - Called peripherals
  - Used to input and output instructions and data

- **Arithmetic and Logic Unit**
  - – Performs arithmetic operations (addition, subtraction)
  - – Performs logical operations (AND, OR, XOR, SHIFT, ROTATE)

COMPUTER SYSTEM COMPONENTS

- **Control Unit**
- – Coordinates the operation of the computer

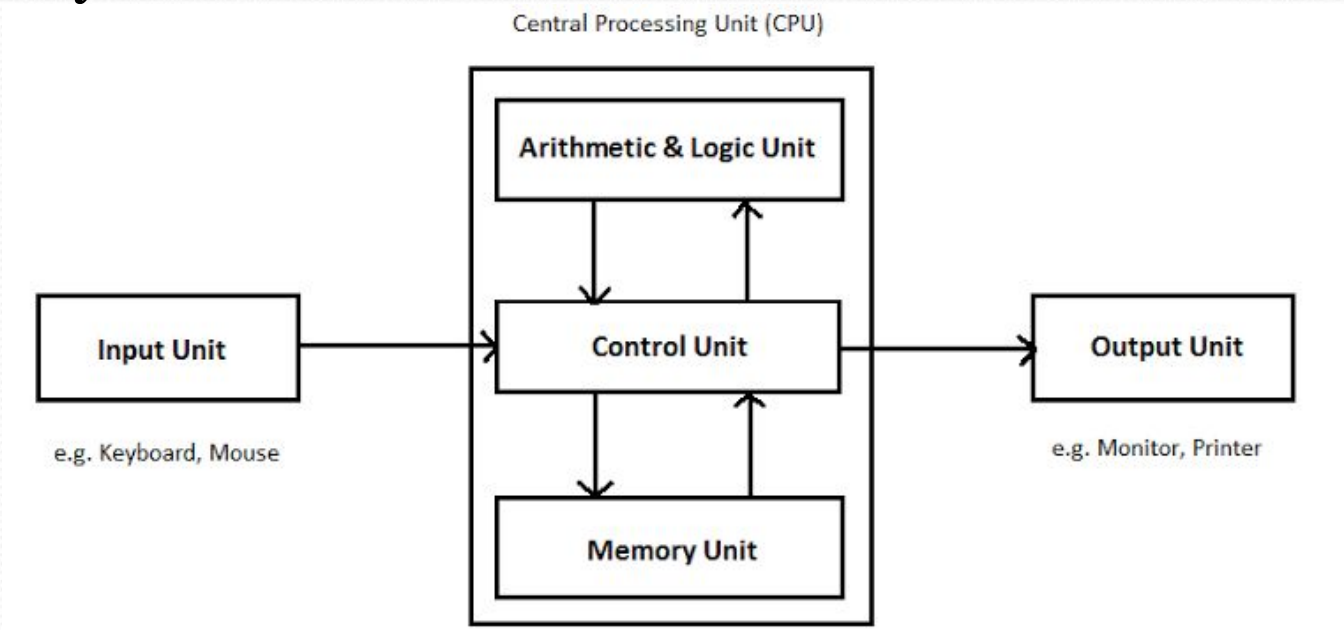
System Interconnection and Interaction

- Bus—A group of lines used to transfer bits between the microprocessor and other components of the computer system. Bus is used to communicate between parts of the computer. There is only one transmitter at a time and only the addressed device can respond.
- Types
  - » Address
  - » Data
  - » Control signals

CPU COMPONENTS

Registers

- Hold data, instructions, or other items.
- Various sizes.
- Program counter and memory address registers must be of same size/width as address bus.
- Registers which hold data must be of same size/width as memory words.



CPU COMPONENTS

Control Unit

- Generates control signals which are necessary for execution of an instruction.
- Connect registers to the bus.
- Controls the data flow between CPU and peripherals (including memory).
- Provides status, control & timing signals required for the operation of memory and I/O devices to the system.
- Acts as a brain of computer system

All actions of the control unit are associated with the decoding and executions of instructions (fetch and

CPU COMPONENTS

Arithmetic and Logic Unit

- Executes arithmetic and logical operations.
- Accumulator is a special 8-bit register associated with ALU.Register ‘A’ in 8085 is an accumulator.
  - Source of one of the operands of an arithmetic or logical operation.
  - serves as one input to ALU.
- Final result of an arithmetic or logical operation is placed in accumulator.

ARITHMETIC AND LOGIC UNIT

ALU performs the following arithmetic & logical operations:

- Addition
- Subtraction
- Logical AND
- Logical OR
- Logical EXCLUSIVE OR
- Complement(logical NOT)
- Increment (add 1)
- Decrement (subtract 1)
- Left shift, Rotate Left, Rotate right
- Clear etc.

- Data and instructions are stored in a single set o f read-write memory. Contents of memory are addressable by memory address, without regard to the type of data contained.Execution occurs in a sequential fashion, unless explicitly altered from one instruction to the other

Von Neumann vs Harvard architecture

	Von Neumann Architecture	Harvard Architecture
Flexibility	High level of flexibility as the memory is shared between instructions and data so the level assigned to each can fluctuate depending on task	Limited flexibility as there is only a certain amount of memory that can be used for data and a certain amount for instructions.
Speed	Speed is limited when compared to harvard due to only having one memory location and set of buses	Two sets of memory and buses mean data can be handled more quickly which would result in decreasing execution time
Examples	Typically used in general purpose computers that will be used for many different purposes.	Typically embedded systems like washing machines, burglar alarms etc.

Performance metrics of CPU:

Short response time for a given piece of work

High throughput (rate of processing work)

Low utilization of computing resource(s)

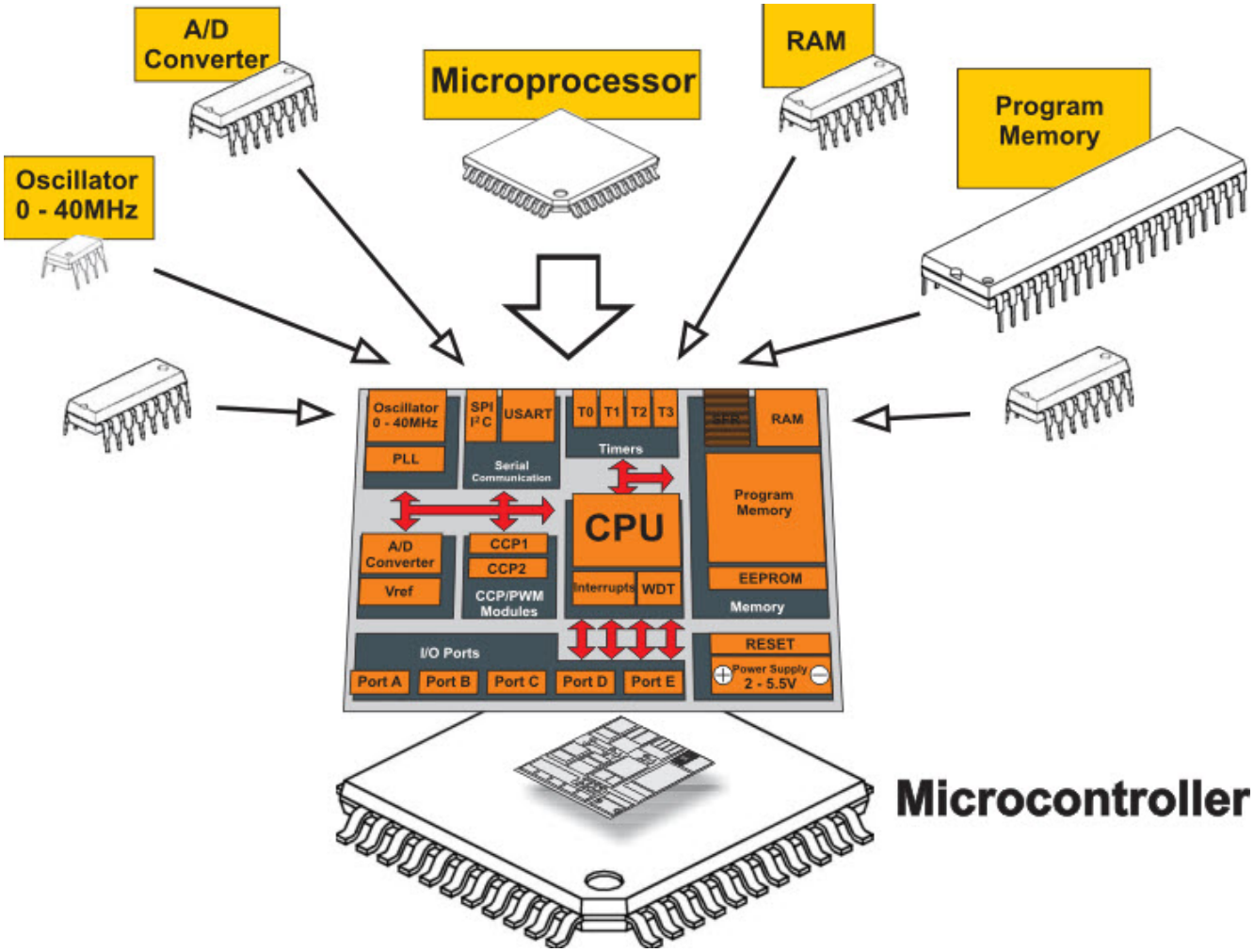
High availability of the computing system or application

Fast (or highly compact) data compression and decompression

High bandwidth

Short data transmission time.

Microprocessor and microcontrollers:



https://www.youtube.com/watch?v=_Cdf68NMTZ0.

Microprocessor	Introduction Date	Introduction Speed	Process Technology	Transistor Count	Addressable Memory	Bits
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8008	Apr, 1972	200 kHz	10,000nm	3,500	16 KB	8
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8085	Mar, 1976	4.77 MHz	3,000nm	6,500	64 KB	8
8086	Jun, 1978	4.77 MHz	3,000nm	29,000	1 MB	16
80286	Feb, 1982	6 MHz	1,500nm	134,000	16 MB	16
80386	Oct, 1985	16 MHz	1,500nm	275,000	4 GB	32
80486	Apr, 1989	25 MHz	1,000nm	1.2 Million	4 GB	32
Pentium	Mar, 1993	60 MHz	800nm	3.1 Million	4 GB	32
Pentium Pro	Nov, 1995	150 MHz	600nm	5.5 Million*	64 GB~	32
Pentium II	May, 1997	233 MHz	350nm	7.5 Million*	64 GB~	32
Pentium III	Feb, 1999	450 MHz	250nm	9.5 Million*	64 GB~	32
Pentium 4	Nov, 2000	1.4 GHz	180nm	42 Million	64 GB~	32

MICROPROCESSOR

- is a semiconductor device consisting of electronic logic circuits
- manufactured by using various fabrication schemes
- capable of performing computing functions
- capable of transporting data/information
- can be divided into 3 segments:
  - Arithmetic and Logic Unit
  - Register Unit
  - Control Unit

History: https://spectrum.ieee.org/the-surprising-story-of-the-first-microprocessors

DIFFERENTIATING FEATURES

FEATURES	MICROPROCESSOR	MICROCONTROLLER
Function	**Process** the **general** task, only	Both **Process** and **Control** the **specific** task
Memory	**No** in-built memory	In-built ROM and RAM memories
Application	**General** purpose (eg. PC, modems, calculators, LED display boards)	**Specific** purpose (eg. A.C. and washing machine, home automation)
Complexity	**More** complex, **large** no. of instructions	**Less** complex, **less** no. of instructions
Cost	**High** (design time is more)	**Low** (design time is low)
Architecture	**Von Neumann** (program and data stored in same memory)	**Harvard** (program and data stored in different memory)
Example	80**85**, 80**86**	80**51**

Microprocessor vs Microcontroller

Memory	Needs external memory & storage, flexible	On-chip memory (RAM, ROM, EEPROM)
Peripheral 	Need additional part via external bus, ideal for complex, powerful applications.	Integrated peripheral SPI, UART, RTI, ADC for direct HW control
Clock Speed 	Fast GHz for heavy duty tasks 	KHz - MHz, efficient for specific tasks. 
Power Consumption 	High power consumption, no low power option. 	Low power consumption with power saving options 
Connectivity	Handles high-speed transfer	supports low to moderate speed communication I2C, SPI, UART
Operating System 	Need full-fledge OS (Linux, WIndows)	OS is optional (no-OS or light weight RTOS)
Usecase	For generic computing, high complexity capacity system	For compact systems, battery powered, or logic processing devices
Cost	Higher cost — suited for performance-driven applications	Lower cost — economical for mass production.

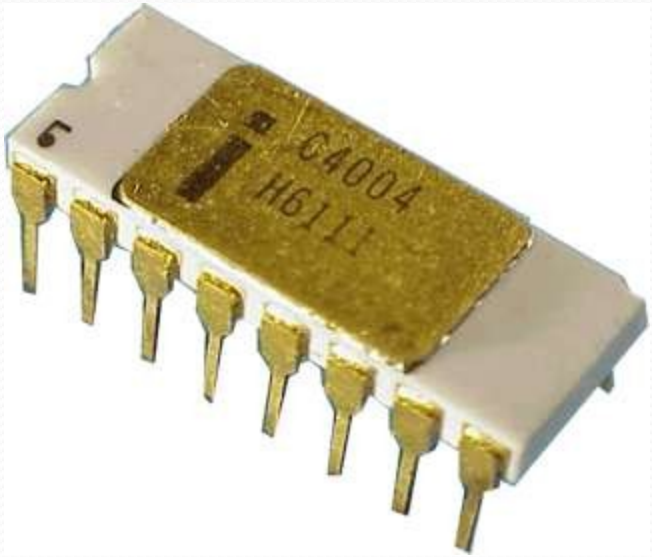
HISTORY OF INTEL MICROPROCESSORS

- Introduction
- 4-Bit Microprocessors
- 8-Bit Microprocessors
- 16-Bit Microprocessors
- 32-Bit Microprocessors
- 64-Bit Microprocessors

4-BIT MICROPROCESSORS

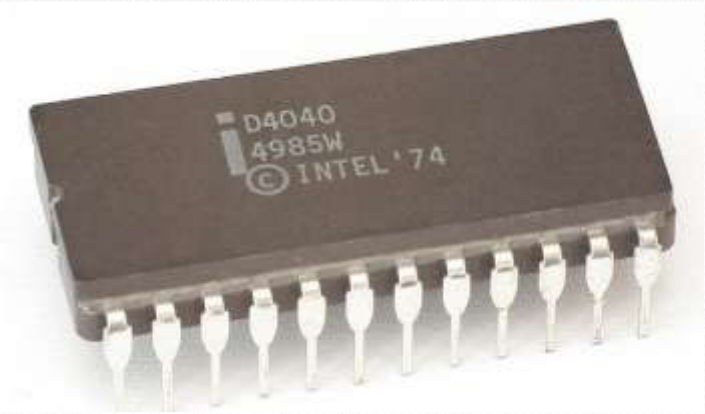
HISTORICAL BACKGROUND OF INTEL MICROPROCESSORS

INTEL 4004



- Introduced in 1971 and noted as 1Generation µP.
- It was the first microprocessor by Intel.
- It was a 4-bit µP.
- Its clock speed was 740KHz.
- It had 2,300 transistors.
- It could execute around 60,000 instructions per second.
- 4 KB main memory
- 45 instructions
- PMOS technology
- It was first programmable device which was used in calculators

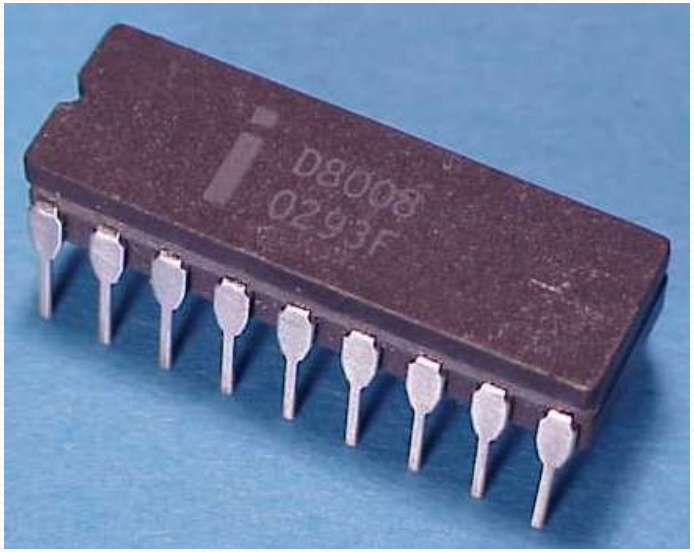
4040



- Introduced in 1974.
- It was also 4-bit µP.

8-BIT MICROPROCESSORS

INTEL 8008



- Introduced in 1972.
- It was first 8-bit μP.
- Its clock speed was 500 KHz.
- Could execute 50,000 instructions per second.

- 8-bit version of 4004
- 16 KB main memory
- 48 instructions
- PMOS technology
- Slow

INTEL 8080



- Introduced in 1974.
- It was also 8-bit µP.
- Its clock speed was 2 MHz.
- It had 6,000 transistors.
- Was 10 times faster than 8008.
- Could execute 5,00,000 instructions per second.

- 64 KB main memory
- 10X faster than 8008
- NMOS technology
- Drawback was that it needed three power supplies.
- Small computers (Microcomputers) were designed in mid 1970's using 8080 as CPU.

INTEL 8085

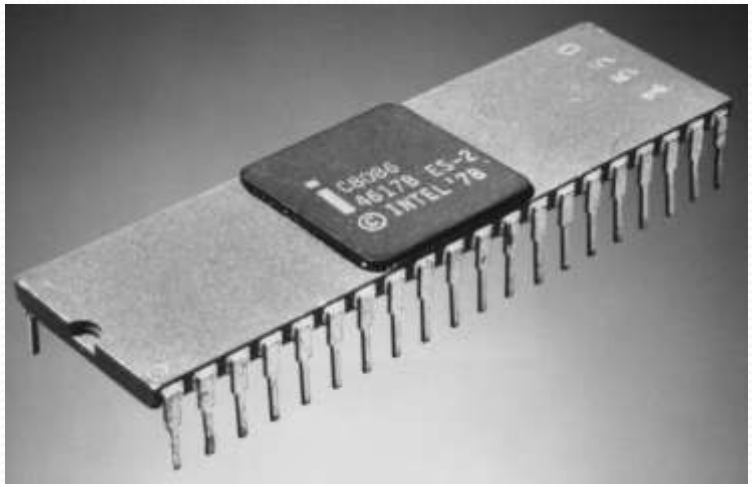


- Introduced in 1976.
- It was also 8-bit µP.
- Its clock speed was 3 MHz.
- Its data bus is 8-bit and address bus is 16-bit.
- It had 6,500 transistors.
- Could execute 7,69,230 instructions per second.
- It could access 64 KB of memory.
- It had 246 instructions.
- Over 100 million copies were sold.

- 64 KB main memory
- NMOS technology
- Uses only one +5v power supply.

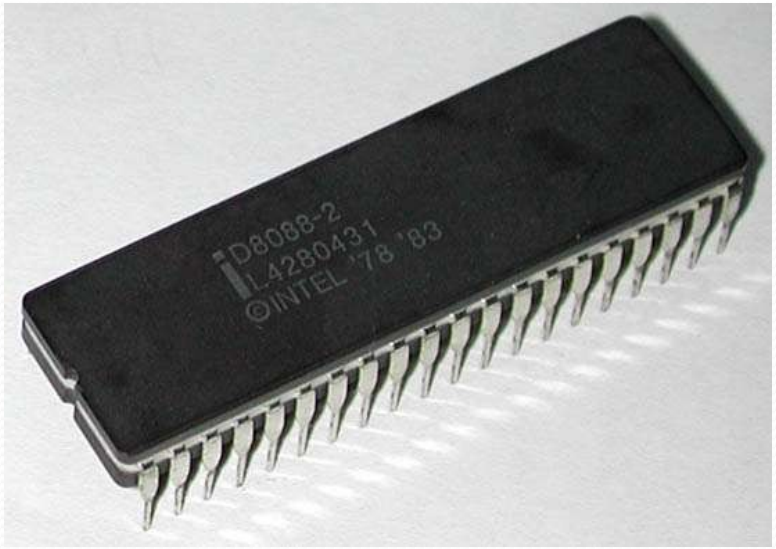
16-BIT MICROPROCESSORS

INTEL 8086



- Introduced in 1978.
- It was first 16-bit µP and noted as 3G.
- Its clock speed is 4.77 MHz, 8 MHz and 10 MHz, depending on the version.
- Its data bus is 16-bit and address bus is 20-bit.
- It had 29,000 transistors and HMOS technology.
- Could execute 2.5 million instructions per second.
- It could access 1 MB of memory.
- It had 22,000 instructions.
- It had ***Multiply*** and ***Divide*** instructions.

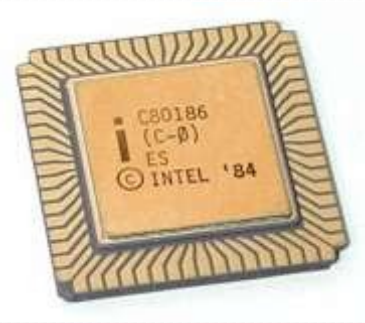
INTEL 8088

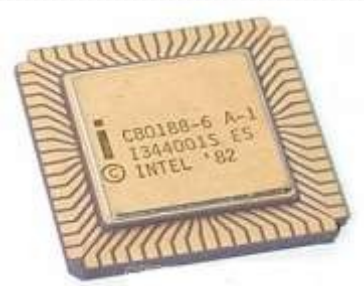


- Introduced in 1979.
- It was also 16-bit μP.
- It was created as a cheaper version of Intel's 8086.
- It was a 16-bit processor with an 8-bit external bus.
- Could execute 2.5 million instructions per second.
- This chip became the most popular in the computer industry when **IBM used it for its first PC**

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INTEL 80186 & 80188





- Introduced in 1982.
- They were 16-bit µPs.
- Clock speed was 6 MHz.
- 80188 was a cheaper version of 80186 with an 8- bit external data bus.
- They had additional components like:
  - Interrupt Controller
  - Clock Generator
  - Local Bus Controller
  - Counters

INTEL 80286



- Introduced in 1982.
- It was 16-bit μP.
- Its clock speed was 8 MHz.
- Its data bus is 16-bit and address bus is 24-bit.
- It could address 16 MB of memory.
- It had 1,34,000 transistors.
- It could execute 4 million instructions per second.

32-BIT MICROPROCESSORS

INTEL 80386



- Introduced in 1986.
- It was first 32-bit µP.
- Its data bus is 32-bit and address bus is 32-bit.
- It could address 4 GB of memory.
- It had 2,75,000 transistors.
- Its clock speed varied from 16 MHz to 33 MHz depending upon the various versions.
- Different versions:
  - 80386 DX
  - 80386 SX
  - 80386 SL
- Intel 80386 became the best selling microprocessor in history.

INTEL 80486



- Introduced in 1989.
- It was also 32-bit μP.
- It had 1.2 million transistors.
- Its clock speed varied from 16 MHz to 100 MHz depending upon the various versions.
- It had five different versions:
  - 80486 DX
  - 80486 SX
  - 80486 DX2
  - 80486 SL
  - 80486 DX4
- 8 KB of cache memory was introduced.

Intel Pentium



- Introduced in 1993.
- It was also 32-bit μP.
- It was originally named 80586.
- Its clock speed was 66 MHz.
- Its data bus is 32-bit and address bus is 32-bit.
- It could address 4 GB of memory.
- Could execute 110 million instructions per second.
- Cache memory:
  - 8 KB for instructions.
  - 8 KB for data.

Intel Pentium Pro



- Introduced in 1995.
- It was also 32-bit µP.
- It had L2 cache of 256 KB.
- It had 21 million transistors.
- It was primarily used in server systems.
- Cache memory:
  - 8 KB for instructions.
  - 8 KB for data.
- It had L2 cache of 256 KB.

INTEL PENTIUM II



- Introduced in 1997.
- It was also 32-bit μP.
- Its clock speed was 233 MHz to 500 MHz.
- Could execute 333 million instructions per second.
- MMX technology was supported.
- L2 cache & processor were on one circuit.

Intel Pentium II Xeon



- Introduced in 1998.
- It was also 32-bit µP.
- It was designed for servers.
- Its clock speed was 400 MHz to 450 MHz.
- L1 cache of 32 KB & L2 cache of 512 KB, 1MB or 2 MB.
- It could work with 4 Xeons in same system.

Intel Pentium III



- Introduced in 1999.
- It was also 32-bit µP.
- Its clock speed varied from 500 MHz to 1.4 GHz.
- It had 9.5 million transistors.

Intel Pentium IV

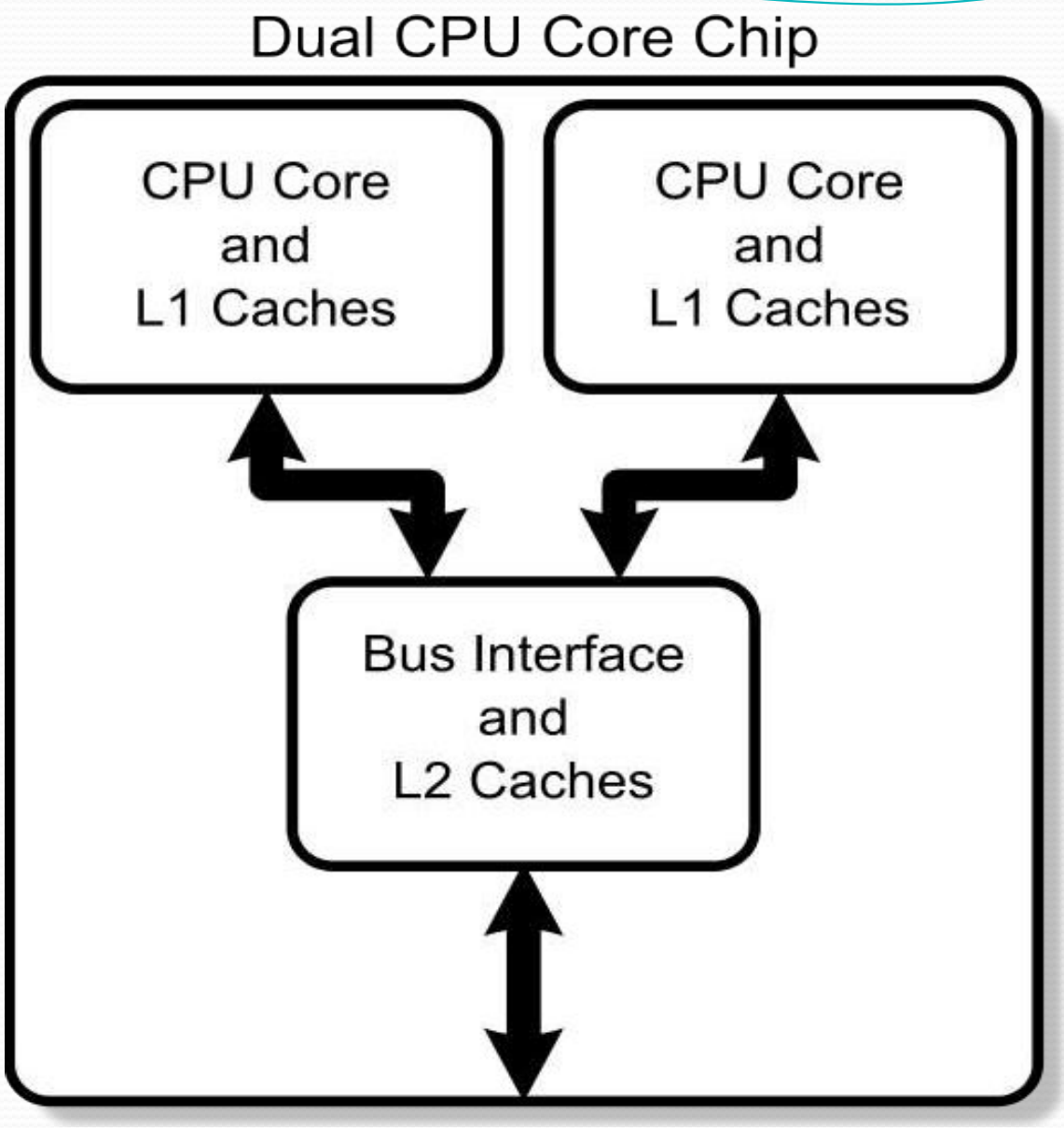


- Introduced in 2000.
- It was also 32-bit μP.
- Its clock speed was from 1.3 GHz to 3.8 GHz.
- L1 cache was of 32 KB & L2 cache of 256 KB.
- It had 42 million transistors.
- All internal connections were made from aluminium to copper.

INTEL DUAL CORE



- Introduced in 2006.
- It is 32-bit or 64-bit µP.
- It has two cores.
- Both the cores have there own internal bus and L1 cache, but share the external bus and L2 cache.
- It supported SMT technology.
- SMT: Simultaneously Multi-Threading
- E.g.: Adobe Photoshop supported SMT.


Dual CPU Core Chip
CPU Core
and
L1 Caches
CPU Core
and
L1 Caches
Bus Interface
and
L2 Caches

64-BIT MICROPROCESSORS

Intel Core 2



- Introduced in 2006.
- It is a 64-bit μP.
- Its clock speed is from 1.2 GHz to 3 GHz.
- It has 291 million transistors.
- It has 64 KB of L1 cache per core and 4 MB of L2 cache.
- It is launched in three different versions:
  - Intel Core 2 Duo
  - Intel Core 2 Quad
  - Intel Core 2 Extreme

Intel Core i3



- Introduced in 2010.
- It is a 64-bit µP.
- It has 2 physical cores.
- Its clock speed is from 2.93 GHz to 3.33 GHz.
- It has 781 million transistors.
- It has 64 KB of L1 cache per core, 512 KB of L2 cache and 4 MB of L3 cache.

Intel Core i5



- Introduced in 2009.
- It is a 64-bit μP.
- It has 4 physical cores.
- Its clock speed is from 2.40 GHz to 3.60 GHz.
- It has 781 million transistors.
- It has 64 KB of L1 cache per core, 256 KB of L2 cache and 8 MB of L3 cache.

Intel Core i7



- Introduced in 2008.
- It is a 64-bit µP.
- It has 4 physical cores.
- Its clock speed is from 2.66 GHz to 3.33 GHz.
- It has 230 billion transistors.
- It has 64 KB of L1 cache per core, 256 KB of L2 cache and 8 MB of L3 cache.

Intel Core i9



- Introduced in 2017.
- It is a 64-bit μP.
- It has 10 physical cores.
- Its clock speed is from 3.33 GHz to 5.2 GHz.
- It has 430 billion transistors.
- It has 64 KB of L1 cache per core, 1 MB of L2 cache per core and 13.75 MB of L3 cache.

Microprocessor	Introduction Date	Introduction Speed	Process Technology	Transistor Count	Addressable Memory	Bits
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Leading MPU Suppliers ($M)

2012 Rank	Company	2011	2012	Percent Change	Percent Marketshare	Main Product Lines
1	Intel	37,435	36,892	-1%	65.3%	x86 PC, server MPUs
2	Qualcomm	4,152	5,322	28%	9.4%	ARM mobile app processors
3	Samsung (+Apple)*	2,614	4,664	78%	8.2%	ARM mobile app processors
4	AMD	4,552	3,605	-21%	6.4%	x86 PC, server MPUs
5	Freescale	1,210	1,070	-12%	1.9%	ARM and embedded MPUs
6	Nvidia	591	764	29%	1.4%	ARM mobile app processors
7	TI	510	565	11%	1.0%	ARM mobile app processors
8	ST-Ericsson**	660	540	-18%	1.0%	ARM mobile app processors
9	Broadcom	295	345	17%	0.6%	ARM mobile app processors
10	MediaTek	280	325	16%	0.6%	ARM mobile app processors

*Includes Apple's custom processors made by Samsung's foundry business.

Source: IC Insights

**Cellphone IC joint venture to be dissovled by STMicroelectronics and Ericsson by 3Q13.

Assembly language, high level language, low level language, machine language:

http://www.itrelease.com/2018/07/difference-between-assembly-language-and-high-level-language/.

MOV A, M : **Assembly language**

7E (0111 1110): **Machine language**

- Assembly language and machine language are treated as low level languages.

- **Assembly language vs high-level language :**

- In assembly language programs written for one processor will not run on another type of processor. In high-level language programs run independently of processor type.
- Performance and accuracy of assembly language code are better than a high-level.
- High-level languages have to give extra instructions to run code on the computer.
- Code of assembly language is difficult to understand and debug than a high-level.
- One or two statements of high-level language expand into many assembly language codes.

- Assembly language can communicate better than a high-level. Some type of hardware actions can only be performed by assembly language.
- In assembly language, we can directly read pointers at a physical address which is not possible in high-level
- Working with bits is easier in assembly language.
- **Assembler** is used to translate code in assembly language while the **compiler** is used to compile code in the high-level.
- The **executable code** of high-level language is larger than assembly language code so it takes a longer time to execute.
- Due to long executable code, high-level programs are less efficient than assembly language programs.
- High-level language programmer does not need to know details about hardware like **registers** in the processor as compared to assembly programmers.
- The most high-level language code is first automatically converted into **assembly code**.

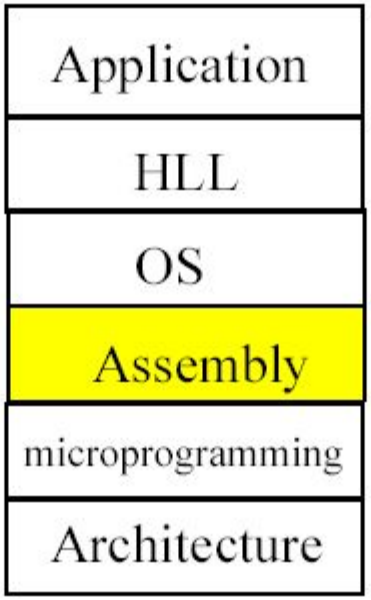
Assembly Programming

- Machine Language
  - binary
  - hexadecimal
  - machine code or object code
- Assembly Language
  - mnemonics
  - assembler
- High-Level Language
  - Pascal, Basic, C
  - compiler

Assembly Language Programming

Motivations

- Why do you learn assembly language?



Microprocessor development tools:

https://csenotesforyou.blogspot.com/2016/12/assembly-language-program-development.html.

Software Tools !

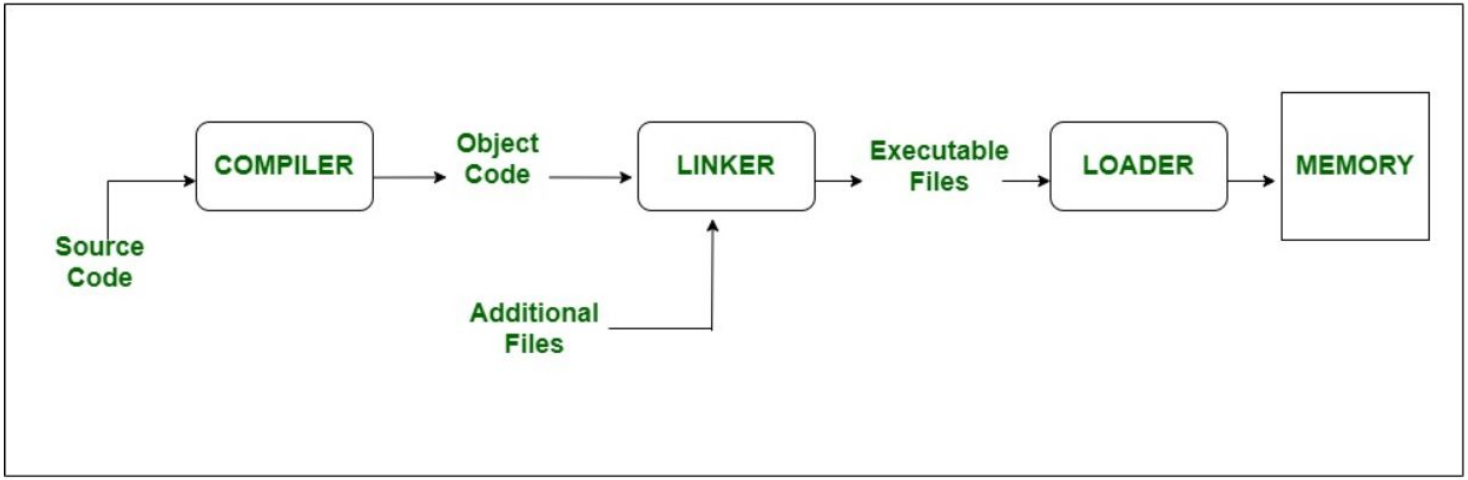
Assembler !
Linker !
Loader !
Compiler !
Libraries !
Simulator !
Debugger !
Locator !

Hardware Tools :

In Circuit Emulator (ICE)
Logic Analyzer
Emulator

- A debugger is a computer program used by programmers to test and debug a target program. Debuggers may use instruction-set simulators, rather than running a program directly on the processor to achieve a higher level of control over its execution. This allows debuggers to stop or halt the program according to specific conditions. However, use of simulators decreases execution speed. When a program crashes, debuggers show the position of the error in the target program. Most debuggers also are capable of running programs in a step-by-step mode, besides stopping on specific points. They also can often modify the state of programs while they are running.

Linker : A linker is special program that combines the object files, generated by compiler/assembler, and other pieces of codes to originate an executable file have. exe extension. In the object file, linker searches and append all libraries needed for execution of file. It regulates the memory space that will hold the code from each module. It also merges two or more separate object programs and establishes link among them.

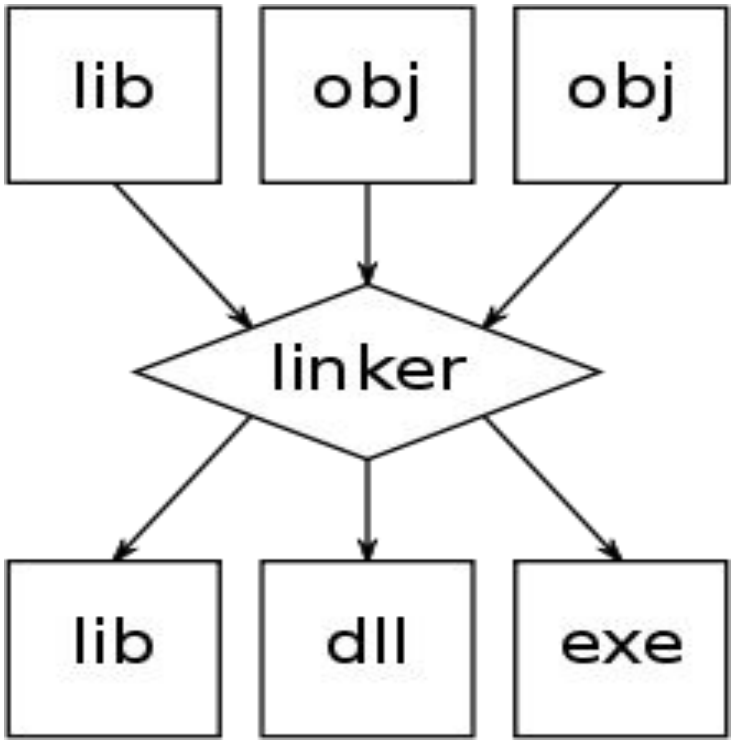


Loader : The loader is special program that takes input of object code from linker, loads it to main memory, and prepares this code for execution by computer. Loader allocates memory space to program. Even it settles down symbolic reference between objects. It is in charge of loading programs and libraries in operating system.

Locator - A locator is a program used to assign the specific addresses of where the segments of object code are to be loaded into memory.

Linking and relocation:

In computing, a **linker** or **link editor** is a computer utility program that takes one or more object files generated by a compiler and combines them into a single executable file, library file, or another 'object' file.



- Computer programs typically are composed of several parts or modules; these parts/modules need not all be contained within a single object file, and in such cases refer to each other by means of symbols as addresses into other modules, which are mapped into memory addresses when linked for execution. For most compilers, each object file is the result of compiling one input source code file.

- When a program comprises multiple object files, the linker combines these files into a unified executable program, resolving the symbols as it goes along. Linkers can take objects from a collection called a *library* or runtime library. Most linkers do not include the whole library in the output; they include only the files that are referenced by other object files or libraries. Library linking may thus be an iterative process, with some modules included requiring additional modules to be linked, and so on.

- **Relocation** is the process of assigning load addresses for position-dependent code and data of a program and adjusting the code and data to reflect the assigned addresses. Prior to the advent of multi process systems, and still in many embedded systems the addresses for objects were absolute starting at a known location, often zero. Since multiprocessing systems dynamically link and switch between programs it became necessary to be able to relocate objects using position-independent code.

- A linker usually performs relocation in conjunction with **symbol resolution**, the process of searching files and libraries to replace symbolic references or names of libraries with actual usable addresses in memory before running a program. Relocation is typically done by the linker at link time, but it can also be done at load time by a relocating loader, or at run time by the running program itself. Some architectures avoid relocation entirely by deferring address assignment to run time; this is known as zero address arithmetic.

- After linking, there has to be re-allocation of the sequences of placing the codes before actually placement of the codes in the memory. The loader program performs the task of reallocating the codes after finding the physical RAM addresses available at a given instant.

Debugger: - A debugger is a program which allows you to load your object code program into system memory, execute the program and troubleshoot or debug it.

The debugger allows you to look at the contents of registers and memory locations after your program runs.

- It allows you to change the contents of registers and memory locations and re-run the program.

- Some debuggers allow you to stop execution after each instruction so that you can check or alter after each register contents.

- A debugger also allows you to set a breakpoint at any point in your program. If you insert a breakpoint at any point in your program, the debugger will run the program up to the instruction where you put the breakpoint and then stop the execution.

Emulator: - An emulator is a mixture of hardware and software.

- It is used to test and debug the hardware and software of an external system, such as the prototype of a microprocessor based instrument. Part of the hardware of an emulator is a multi wire cable which connects the host system to the system being developed. Ex: FPGAs, USRPs.

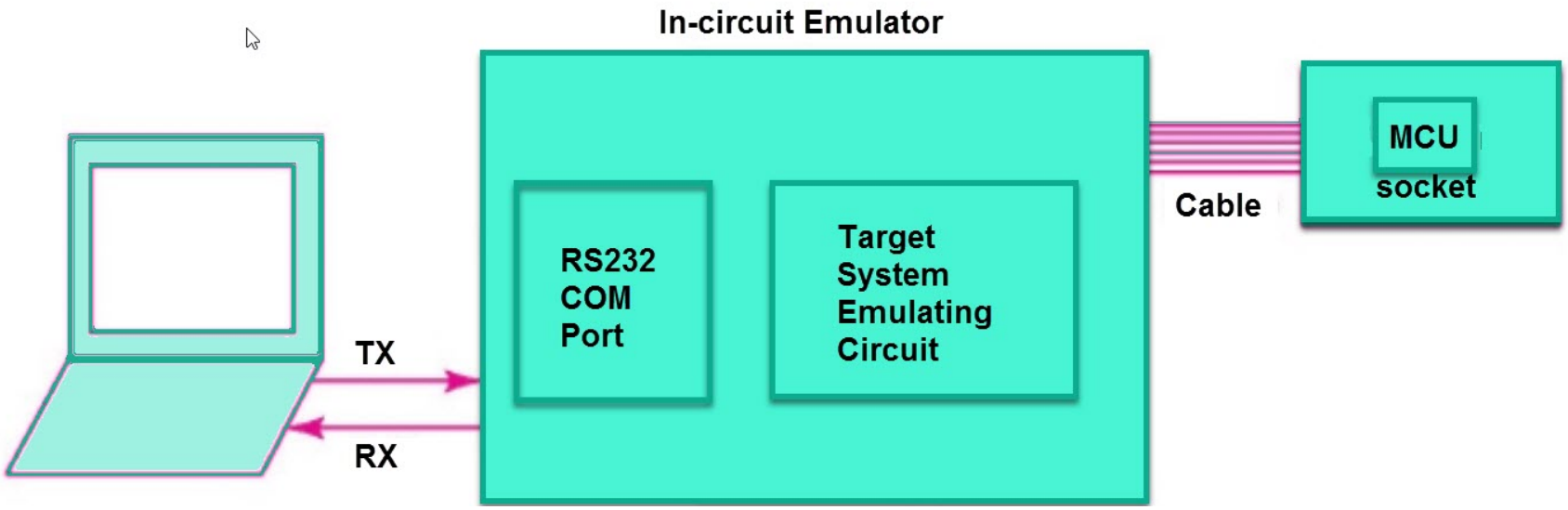
Debugging, Simulator, Emulator, In-circuit emulator:

https://www.youtube.com/watch?v=4wmDsd53ibE.

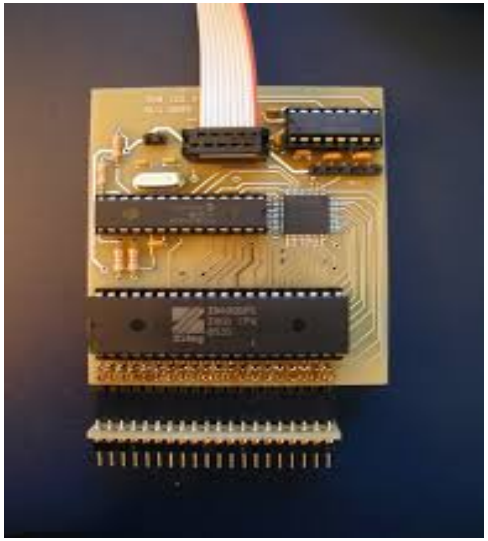
ICE:

Embedded systems pose unique debugging challenges. With neither terminal nor display (in most cases), there's no natural way to probe these devices, to extract the behavioural information needed to find what's wrong. They let us connect an external computer to the system being debugged to enable single stepping, breakpoints, and all of the debug resources enjoyed by programmers of desktop computers.

In-circuit emulation (**ICE**) is the use of a hardware device or **in-circuit emulator** used to debug the software of an embedded system. It operates by using a processor with the additional ability to support debugging operations, as well as to carry out the main function of the system. Particularly for older systems, with limited processors, this usually involved replacing the processor temporarily with a hardware emulator: a more powerful although more expensive version. It was historically in the form of bond-out processor which has many internal signals brought out for the purpose of debugging. These signals provide information about the state of the processor.

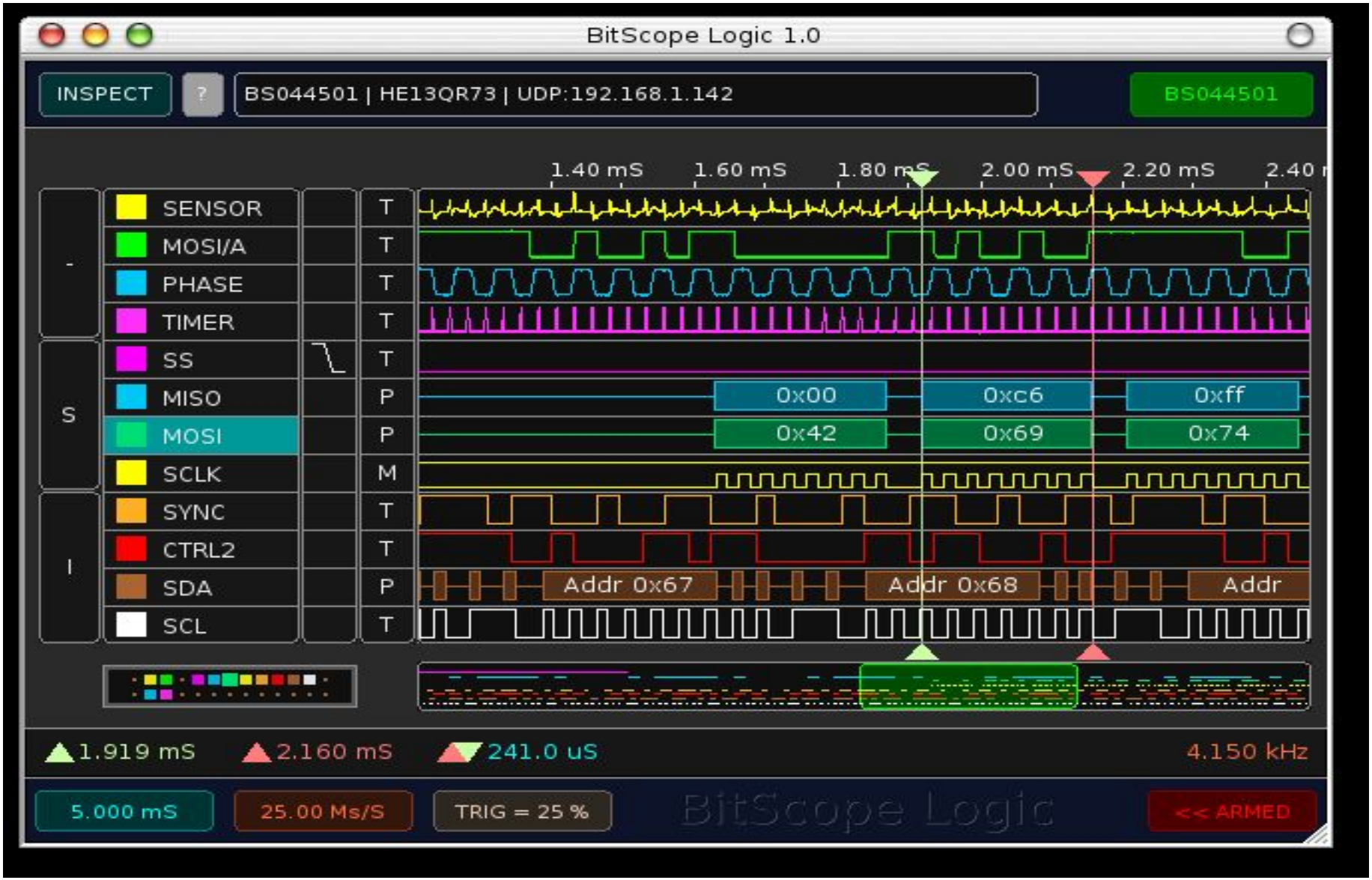


- An in-circuit emulator (ICE) provides a window into the embedded system. The programmer uses the emulator to load programs into the embedded system, run them, step through them slowly, and view and change data used by the system's software.

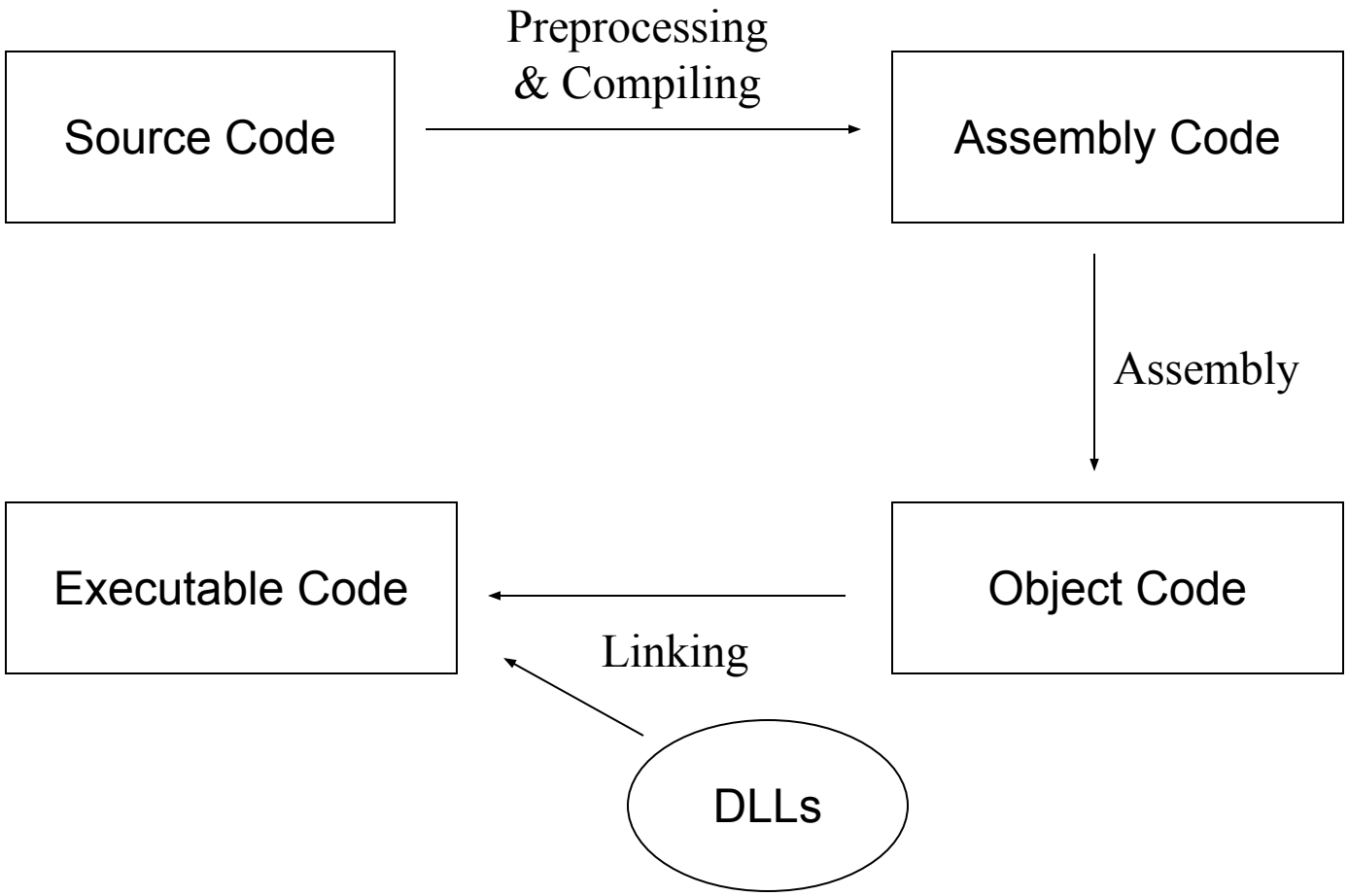


Logic analysers:

https://www.radio-electronics.com/info/t_and_m/logic_analyzer/logic_analyzer.php.



WHAT DOES IT MEAN TO DISASSEMBLE CODE?



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